

Aryan Prodduturi

Software Developer

EDUCATION

Georgia Institute of Technology, Atlanta — BSCS, 2026

- Concentrations: Theory and Modeling & Simulation
- Relevant Coursework: Algorithms, Data Structures, Systems & Networks, Probability & Statistics
- Graduate with Highest Honors

EXPERIENCE

Fullstack Developer, Sri Radha's Veg Cuisine — 2025

- Designed and built a **centralized digital menu** display system, enabling real-time content updates.
- Developed food waste tracking and **forecasting tools** to analyze consumption patterns.
- Implemented a **SQL-backed** management system to track daily menus, items, and food wastage data.
- Owned **end-to-end development**, including database design, API integration, UI/UX design, and deployment.

Website Manager, Arkaya Yoga Studio — 2025

- Led website updates following a brand re-launch, improving **SEO** structure, page performance, and user experience.
- **Automated** blog publishing workflows on **WordPress** and **custom scripts** to streamline content updates.
- Built an online **appointment scheduling system** that reduced manual work and improved booking efficiency.

High School Math Teacher, Khan Lab School — 2024

- Designed and taught an **accredited**, project-based geometry course aligned with **state and institutional standards**.
- Applied **mastery-based** assessment techniques to track student progress and ensure **conceptual understanding**.
- Achieved a 100% A-grade completion rate while maintaining rigorous academic expectations.

Programming Intern, Big Data Trunk — 2021

- Developed a Python-based **data analysis** toolkit to generate **key metrics** and **visualizations** from datasets.
- Improved data processing efficiency by standardizing analysis workflows and reusable components.
- Collaborated with a distributed engineering team using Agile/SCRUM methodologies.
- Built reusable **Python modules** for data ingestion, transformation, and visualization.

PROJECTS

A small selection of my 50+ Projects: <https://github.com/laryanpro>

Bayesian NBA Player Projection Model (Python)

- Built a **two-stage forecasting pipeline** that predicts players stats conditioned on predicted playing time.
- Implemented Bayesian hierarchical regression with **partial pooling** across opponents and game contexts to generate probabilistic forecasts under sparse data.
- Designed a **walk-forward backtesting** framework and evaluated out-of-sample performance via forward testing.

Quantitative Trading Algorithm (Javascript)

- Implemented indicator calculations and strategy backtesting using live market APIs.
- Achieved **15% ROI over 6 months** in forward testing on daily timeframes.

Restaurant Waste Management System (JS, Svelte, SQL)

- Developed a waste tracking system that records daily prepared and leftover quantities for individual buffet items.
- Designed and implemented a **structured data pipeline** for collecting and querying waste and preparation data.
- Generated **analytical summaries** to identify trends and inefficiencies in food production.

Intro to Computer Science Course

- Independently constructed an introductory level **computer science course** for 11th and 12th graders at a High School
- **Accredited the course** in the A-G requirements meaning students received college transfer credits
- Used Mastery Based Learning and Project Based Learning to teach a practically applicable course

Text-Based Physics Simulator (C++, Shell Script)

- Implemented a C++ physics engine supporting projectile and rotational dynamics with physics coordinates mapped to non-square terminal character grids for accurate spatial rendering.
- Built an **adaptive rendering loop** in ncurses that adjusts simulation speed to match real-time frame rate.
- Exposed a **runtime API** for live manipulation of physics objects, enabling dynamic adjustment of forces, positions, and simulation parameters without restarting.

SKILLS

- **Languages:** Python, JavaScript, C/C++, Java, R, Lua, Swift, Shell
- **Frameworks & Tools:** Svelte/SvelteKit, SQL, Git, REST APIs
- **Other:** Adobe Photoshop, Adobe Illustrator, Spreadsheet Modeling